Mini Project: Student Feedback Analytics for Educational Institute

PART 2 — Data Collection via Google Form

Data is collected using a Google Form shared among students. Each response represents one student feedback record. The form takes less than 2 minutes to fill.

Google Form Fields & Settings

#	Question / Field	Type	Validation / Notes
1	Name	Short answer	Required
2	Roll Number	Short answer	Unique roll number
3	Department	Dropdown	Computer / Civil / Mechanical
4	Year	Multiple Choice	First/Second/Third/Final
5	Rate Teaching Quality	Linear Scale	1–5
6	Lab Facilities Rating	Linear Scale	1–5
7	Faculty Support Rating	Linear Scale	1–5
8	Overall Satisfaction	Multiple Choice	Poor/ Average/ Good/ Excellent
9	Suggestions for Improvement	Paragraph	Optional
10	Recommend Institute?	Multiple Choice	Yes / No

PART 3 — Raw Data Collected (Google Sheets Sample)

Form_Responses

Timestamp	Email address	Name	Roll Number	Department	Year	Rate Teaching Quality	Lab Facilities Rating	Faculty Support Rating	Overall Satisfaction	Suggestions for Improvement	Would You Recommend This Institute?
28/04/2026 13:38:31	bhansalitejas7498@gm	Tejas	39	Computer	Third Year	3	4	4	Good		Yes
28/04/2026 13:38:39	yashsurana2005810@g	Yash surana	22	Computer	Third Year	5	4	4	Good	No	Yes
28/04/2026 13:38:58	bhansalitejas7498@gm	Anit	47	Computer	First Year	2	4	3	Average		No
28/04/2026 13:39:20	yashsurana2005810@g	Nirmal chatur	25	Computer	Third Year	4	5	4	Good	No	Yes
28/04/2026 13:39:27	bhansalitejas7498@gm	Suraj	60	Mechanical	Final Year	4	5	1	Average		No
28/04/2026 13:41:24	yashsurana2005810@g	Om	65	Computer	Third Year	4	3	5	Excellent		Yes
28/04/2026 13:41:59	dev264108@gmail.com	Devang Kulkarni	22	Computer	Final Year	4	4	3	Good	.	Yes
28/04/2026 13:42:26	yashsurana2005810@g	Chirag	9	Computer	Third Year	4	4	3	Good		Yes
28/04/2026 13:46:00	rushabhshah2410@gm	Shah Rushabh Abhay	61	Computer	Second Year	5	5	5	Excellent	NA	Yes
28/04/2026 13:46:07	shreyapachore06@gma	Pachore Shreya	52	Computer	Second Year	5	5	5	Excellent	No	Yes
28/04/2026 13:46:14	siddheshnikumb@gmai	Siddhesh Nitin Nikumb	38	Computer	Third Year	5	5	5	Excellent	No	Yes
28/04/2026 13:46:28	absonawane2006@gm	Abhishek Sonawane	67	Computer	Second Year	5	5	5	Excellent		Yes
28/04/2026 13:46:30	prathmeshbagul03@gm	Prathmesh bagul	48	Computer	Final Year	5	5	5	Excellent		No
28/04/2026 13:47:16	siddheshnikumb@gmai	Siddhesh Nitin Nikumb	38	Computer	Third Year	4	4	4	Excellent	No	Yes
28/04/2026 13:47:18	bhansalitejas7498@gm	Sonu	23	Mechanical	Final Year	3	2	4	Average		Yes
28/04/2026 13:47:35	yashsurana2005810@g	Yash kale	66	Mechanical	Third Year	1	4	2	Good		No
28/04/2026 13:47:57	tejasjain.genai@gmail.c	Somya	5	Civil	Third Year	2	1	2	Poor		No
28/04/2026 13:48:17	codeclubsnjbcoe@gma	Chirag Ghansham Shah 09		Computer	Third Year	4	3	3	Average	NA	Yes
28/04/2026 13:48:22	tejasjain.genai@gmail.c	Tej	45	Civil	Second Year	1	1	1	Average		Yes
28/04/2026 13:48:45	tejasjain.genai@gmail.c	Gunjan	45	Mechanical	Final Year	2	3	2	Average		No
28/04/2026 13:48:48	yashsurana67@gmail.co	Karan	9	Mechanical	Final Year	5	4	4	Average		No
28/04/2026 13:49:08	tejasjain.genai@gmail.c	Dvbsan	46	Civil	Third Year	2	2	2	Poor	No	No
28/04/2026 13:49:15	yashsurana67@gmail.co	Daksh	56	Civil	Third Year	3	3	5	Average		Yes
28/04/2026 13:51:20	chiragghanshamshah12	Sanskar shah	10	Computer	First Year	1	1	1	Poor	Djdk	Yes
28/04/2026 13:52:13	chiragshah835@gmail.c	Piyush shah	11	Computer	Second Year	1	2	5	Good	.	Yes
28/04/2026 13:54:49	shreyapachore06@gma	Shreyash Pachore	61	Mechanical	Second Year	3	4	5	Excellent	No	Yes
28/04/2026 13:54:55	aditisinkar@gmail.com	Anuj sinkar	45	Civil	First Year	2	4	1	Excellent	No	No
28/04/2026 13:55:41	omshrawge333@gmail.	Om Deepak Shrawge	52	Computer	Third Year	5	5	5	Excellent	No	Yes
28/04/2026 13:57:48	omshrawge333@gmail.	Moksham Ashish Parek	56	Mechanical	Third Year	5	5	5	Excellent	No	No
28/04/2026 13:59:29	crezotex@gmail.com	Swani	5	Electrical	Second Year	3	3	3	Excellent		Yes
28/04/2026 13:59:34	ugaleshruti4@gmail.cor	Shruti Dhananjay Ugale	72	Computer	Second Year	5	5	5	Excellent		Yes

Google form Response Sheet : DSBDA 2 Compusory

Data Cleaning & Preprocessing

Step 1 — Identify Missing Values

```
import pandas as pd

df=pd.read_csv("student_feedback.csv")

print(df.isnull().sum())
print("Total missing:",df.isnull().sum().sum())
print("Shape:",df.shape)
```

Output :

```
Timestamp                                   0
Email address                               0
Name                                        0
   Roll Number                              0
  Department                                0
  Year                                      0
  Rate Teaching Quality                     0
 Lab Facilities Rating                      0
  Faculty Support Rating                    0
  Overall Satisfaction                      0
  Suggestions for Improvement              17
   Would You Recommend This Institute?      0
dtype: int64
Total missing: 17
Shape: (31, 12)
```

Step 2 — Handle Missing Values

```
df["Rate Teaching Quality"]=df["Rate Teaching Quality"].fillna(
df["Rate Teaching Quality"].mean()
)

df["Lab Facilities Rating"]=df["Lab Facilities Rating"].fillna(
df["Lab Facilities Rating"].mean()
)

df["Overall Satisfaction"]=df["Overall Satisfaction"].fillna(
df["Overall Satisfaction"].mode()[0]
)

df["Suggestions for Improvement"]=df[
"Suggestions for Improvement"
].fillna("No Suggestion")

print("Missing after fix:",df.isnull().sum().sum())
```

Output

```
Missing after fix: 0
```

Step 3 — Remove Duplicate Records

```
print("Duplicates found:",df.duplicated().sum())

df=df.drop_duplicates()

print("Rows after removing duplicates:",len(df))
```

Output:

```
Duplicates found: 0
```

```
Rows after removing duplicates: 31
```

Step 4 — Detect and Handle Outliers (IQR Method)

```
Q1=df["Rate Teaching Quality"].quantile(0.25)
Q3=df["Rate Teaching Quality"].quantile(0.75)

IQR=Q3-Q1

lower=Q1-1.5*IQR
upper=Q3+1.5*IQR

df=df[
(df["Rate Teaching Quality"]>=lower)&
(df["Rate Teaching Quality"]<=upper)
]

print("Rows after outlier removal:",len(df))

print(
"Teaching Rating Range:",
df["Rate Teaching Quality"].min(),
"-",
df["Rate Teaching Quality"].max()
)
```

Output

```
Rows after outlier removal: 31
Teaching Rating Range: 1 - 5
```

Clean Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Timestam	Email add	Name	Roll Numb	Departme	Year	Rate Teac	Lab Facilit	Faculty Su	Overall Sa	Suggestio	Would You	Satisfactic	Scaled_Te
2	#######	bhansalite	Tejas	39	Computer	Third Year	3	4	4	Good		Yes	2	0.5
3	#######	yashsuran	Yash surar	22	Computer	Third Year	5	4	4	Good	No	Yes	2	1
4	#######	bhansalite	Anit	47	Computer	First Year	2	4	3	Average	No Sugges	No	1	0.25
5	#######	yashsuran	Nirmal cha	25	Computer	Third Year	4	5	4	Good	No	Yes	2	0.75
6	#######	bhansalite	Suraj	60	Mechanic	Final Year	4	5	1	Average	No Sugges	No	1	0.75
7	#######	yashsuran	Om	65	Computer	Third Year	4	3	5	Excellent	No Sugges	Yes	3	0.75
8	#######	dev26410	Devang Ku	22	Computer	Final Year	4	4	3	Good	.	Yes	2	0.75
9	#######	yashsuran	Chirag	9	Computer	Third Year	4	4	3	Good	No Sugges	Yes	2	0.75
10	#######	rushabhsh	Shah Rush	61	Computer	Second Ye	5	5	5	Excellent	No Sugges	Yes	3	1
11	#######	shreyapac	Pachore S	52	Computer	Second Ye	5	5	5	Excellent	No	Yes	3	1
12	#######	siddheshn	Siddhesh N	38	Computer	Third Year	5	5	5	Excellent	No	Yes	3	1
13	#######	absonawa	Abhishek S	67	Computer	Second Ye	5	5	5	Excellent	No Sugges	Yes	3	1
14	#######	prathmesl	Prathmesl	48	Computer	Final Year	5	5	5	Excellent	No Sugges	No	3	1
15	#######	siddheshn	Siddhesh N	38	Computer	Third Year	4	4	4	Excellent	No	Yes	3	0.75
16	#######	bhansalite	Sonu	23	Mechanic	Final Year	3	2	4	Average	No Sugges	Yes	1	0.5
17	#######	yashsuran	Yash kale	66	Mechanic	Third Year	1	4	2	Good	No Sugges	No	2	0
18	#######	tejasjain.g	Somya	5	Civil	Third Year	2	1	2	Poor	No Sugges	No	0	0.25
19	#######	codeclubs	Chirag Gh	9	Computer	Third Year	4	3	3	Average	No Sugges	Yes	1	0.75
20	#######	tejasjain.g	Tej	45	Civil	Second Ye	1	1	1	Average	No Sugges	Yes	1	0
21	#######	tejasjain.g	Gunjan	45	Mechanic	Final Year	2	3	2	Average	No Sugges	No	1	0.25
22	#######	yashsuran	Karan	9	Mechanic	Final Year	5	4	4	Average	No Sugges	No	1	1
23	#######	tejasjain.g	Dvbsan	46	Civil	Third Year	2	2	2	Poor	No	No	0	0.25
24	#######	yashsuran	Daksh	56	Civil	Third Year	3	3	5	Average	No Sugges	Yes	1	0.5
25	#######	chiraggha	Sanskar sh	10	Computer	First Year	1	1	1	Poor	Djdk	Yes	0	0
26	#######	chiragshal	Piyush sha	11	Computer	Second Ye	1	2	5	Good	.	Yes	2	0
27	#######	shreyapac	Shreyash F	61	Mechanic	Second Ye	3	4	5	Excellent	No	Yes	3	0.5
28	#######	aaditisinka	Anuj sinka	45	Civil	First Year	2	4	1	Excellent	No	No	3	0.25
29	#######	omshrawg	Om Deepa	52	Computer	Third Year	5	5	5	Excellent	No	Yes	3	1
30	#######	omshrawg	Moksham	56	Mechanic	Third Year	5	5	5	Excellent	No	No	3	1
31	#######	crezotex@	Swani	5	Electrical	Second Ye	3	3	3	Excellent	No Sugges	Yes	3	0.5
32	#######	ugaleshru	Shruti Dha	72	Computer	Second Ye	5	5	5	Excellent	No Sugges	Yes	3	1
33														

Cleaning Dataset: